

Electronics Sessional

(EEE 2188)

Experiment No. 01

Familiarization of Different Equipment
Involved with Electronics Sessional

Department of Electrical & Electronic Engineering

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EXPERIMENT NO. 1

1.1 Objectives

The objectives of this experiment are to learn about the commonly used equipment in the Electronics Sessional, to observe the various parameters of an electric circuit, to identify the different instruments ordinarily employed for measurement, and to understand the use of these instruments and components in experimental work.

1.2 Theory

Electronics is a scientific and engineering discipline that studies and applies the principles of physics in order to design, create, and operate devices capable of manipulating electrons and other electrically charged particles [1]. As a subfield of physics and electrical engineering, it exploits active devices—such as transistors, diodes, and integrated circuits—to control and amplify the flow of electric current, and to convert signals from one form to another: for instance, from alternating current (AC) to direct current (DC), or from analog signals to digital ones.

The evolution of electronics over the last century has been driven by a relentless reduction in component size and cost alongside dramatic improvements in speed, efficiency, and reliability. Beginning with vacuum tubes in the early twentieth century, the field transitioned first to discrete semiconductor devices and then to the densely integrated circuits that now underpin everything from household appliances to satellite communication systems. Each generation of components is governed by the same fundamental physical laws—Ohm’s law, Kirchhoff’s laws, and Maxwell’s equations—yet their practical realization has changed enormously. Understanding the individual components at the heart of any electronic circuit is therefore the essential first step for any engineering student.

In a typical electronics laboratory, a circuit under investigation consists of passive components (resistors, capacitors) that store or dissipate energy, and active components (diodes, transistors, operational amplifiers) that can amplify or switch signals [2]. Measurement instruments such as the multimeter and the oscilloscope allow the engineer to probe voltages, currents, and waveforms at any node of the circuit, while prototyping boards—the breadboard and the veroboard—provide flexible platforms for assembling circuits without committing to a perma-

ment printed-circuit-board (PCB) layout. The soldering iron, finally, is the hand tool by which components are permanently bonded to a PCB once a design has been validated. Together, these components and instruments form the core toolkit of the electronics practitioner, and familiarity with each of them is the primary goal of this experiment.

1.3 Major Components and Equipment

1.3.1 Resistor

Definition. A resistor is a passive, two-terminal electronic component that implements electrical resistance as a circuit element [3]. It obeys Ohm's Law ($V = IR$), meaning the voltage drop across it is directly proportional to the current flowing through it. Resistance is measured in ohms (Ω), and resistors values are most commonly indicated by a system of color-coded bands printed on their body.



Figure 1.1: Resistor and color-band coding chart

Specifications. The primary specifications of a resistor are its resistance value (ranging from fractions of an ohm up to several gigaohms), its tolerance (the permissible deviation from the nominal value, commonly $\pm 1\%$, $\pm 5\%$, or $\pm 10\%$), and its power rating (the maximum heat it can safely dissipate, typically $\frac{1}{8}$ W to 2 W for through-hole types and up to several hundred watts for industrial power resistors) [4]. The temperature coefficient of resistance (TCR), expressed in parts per million per degree Celsius ($\text{ppm}/^\circ\text{C}$), quantifies how much the resistance drifts with temperature; carbon-film resistors exhibit TCR values of ± 100 to ± 500 $\text{ppm}/^\circ\text{C}$, while metal-film types achieve as low as ± 25 $\text{ppm}/^\circ\text{C}$, and precision metal-foil resistors can reach ± 0.2 $\text{ppm}/^\circ\text{C}$ [4]. Resistors are available in through-hole (axial lead) and surface-mount (SMD) packages; SMD types are labeled using a three- or four-digit numeric code rather than color bands.

Purpose of Use. The essential purpose of a resistor is to limit or regulate the flow of electric current in a circuit. Without current-limiting resistors, sensitive components such as LEDs

and transistors would draw destructive levels of current. Beyond simple current limitation, resistors are used to divide voltage (voltage dividers), set the gain of amplifiers, bias the operating point of transistors, terminate transmission lines to prevent signal reflection, and provide pull-up or pull-down functionality on digital logic lines.

Applications. Resistors are ubiquitous in virtually every electronic system. They appear in signal conditioning networks in communication devices, load regulators in power supplies, sensor networks for heat, light, and force sensing (thermistors, LDRs, strain gauges), biasing circuits for BJTs and MOSFETs, and protection networks at the inputs and outputs of integrated circuits [2]. In power electronics, large wirewound or grid-type resistors are used for dynamic braking in motor drives and for load-bank testing of generators.

1.3.2 Multimeter

Definition. A multimeter—also known as a volt-ohm meter (VOM)—is a multi-function electronic measuring instrument that combines the capabilities of a voltmeter, an ammeter, and an ohmmeter in a single handheld or bench-mounted device [5]. Modern digital multimeters (DMMs) display readings on a numeric LCD, whereas the earlier analog multimeter uses a moving-coil mechanism with a pointer deflecting over a graduated scale.

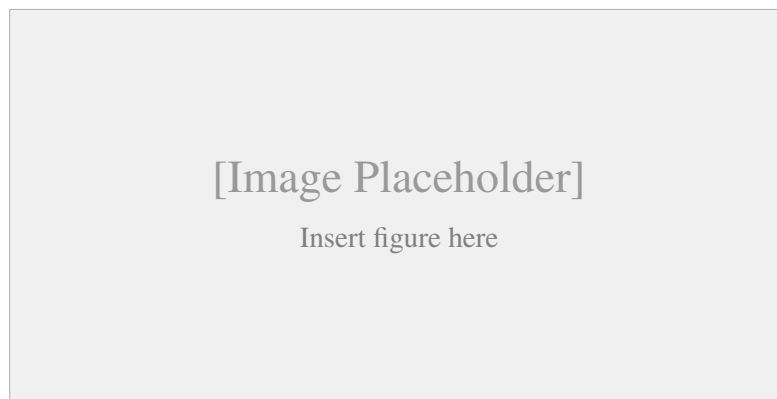


Figure 1.2: Digital multimeter (front view with controls)

Specifications. A typical handheld DMM measures DC voltage in ranges from 200 mV to 600 V, AC voltage up to 600 V, DC current from 200 μ A to 10 A, resistance from 200 Ω to 2 M Ω , and sometimes capacitance and frequency as well [6]. Standard portable DMMs achieve a DC voltage accuracy of $\pm 0.5\%$ of reading, while laboratory-grade bench instruments can attain accuracies better than $\pm 0.01\%$; analog multimeters typically carry a less precise rating of $\pm 3\%$ full-scale deflection [5]. Input impedance on most DMMs is fixed at 10 M Ω , high enough to avoid loading down the circuit under test.

Purpose of Use. The multimeter is used to verify the operating voltages and currents at nodes of a circuit, to check component values (resistance, capacitance), to test diodes and

transistors, and to measure continuity—that is, whether an electrical connection is sound. When used as a voltmeter, it is connected in parallel with the element under test; when used as an ammeter, it must be inserted in series, typically using a low-value shunt resistor and a dedicated 10 A terminal.

Applications. Multimeters are indispensable in electronics laboratories for circuit debugging, component testing, and quality verification. They are equally essential in field service for troubleshooting household wiring, automotive electrical systems, industrial motor drives, and battery-powered consumer devices [5]. Specialized multimeters—sometimes called *clamp meters*—measure AC current through a wire without breaking the circuit, which is invaluable for high-current industrial diagnostics.

1.3.3 Project Board / Breadboard

Definition. A breadboard (or solderless breadboard) is a reusable construction base used to build and test prototype electronic circuits without requiring any soldering [7]. It consists of a perforated plastic block housing an array of spring-clip contacts beneath the holes. Components are inserted into the holes and held by friction, while internal copper strips connect groups of holes to form electrical nodes.

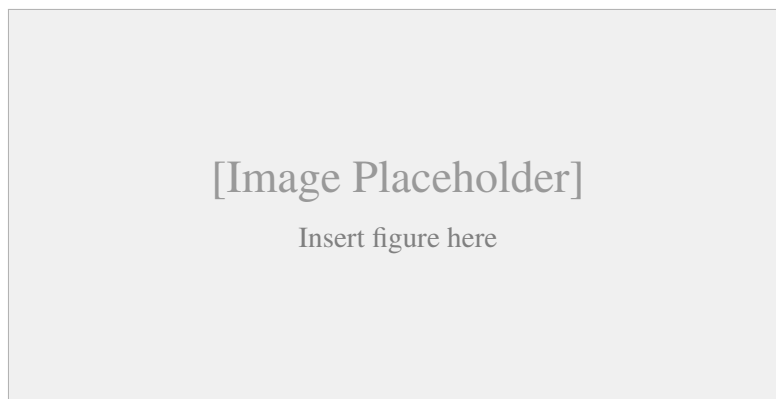


Figure 1.3: Solderless breadboard showing internal connection pattern

Specifications. The holes on a standard breadboard are spaced 0.1 inches (2.54 mm) apart, matching the pin pitch of common dual in-line package (DIP) integrated circuits. A full-size breadboard typically offers 830 tie-points arranged in two terminal strips of 63 rows each and two pairs of power rails running along the edges [7]. Practical breadboards exhibit parasitic capacitance of a few picofarads and contact resistance of a few ohms; these parasitics limit reliable operation to frequencies below approximately 10 MHz [8].

Purpose of Use. The breadboard allows engineers and students to assemble and modify circuits quickly, test different component configurations, and verify circuit behavior—all without the commitment and expense of a permanent soldered PCB. Its reusable nature

makes it particularly suitable for iterative design and experimentation: a component placed incorrectly can be moved to another position in seconds.

Applications. Breadboards are used throughout electronics education, prototyping, and research. They are the standard platform for testing analog filter designs, digital logic circuits, microcontroller-based systems (such as Arduino and Raspberry Pi), and RF circuits at moderate frequencies [8]. Once a design is validated on a breadboard, it is transferred to a permanent PCB or, for intermediate permanence, to a veroboard.

1.3.4 Capacitor

Definition. A capacitor is a passive two-terminal electronic component that stores and releases electrical energy in an electric field [1]. It consists of two conductive plates separated by an insulating material called the dielectric. The ability to store charge is quantified by its capacitance C , defined as $C = Q/V$, where Q is the stored charge and V is the voltage across the plates.

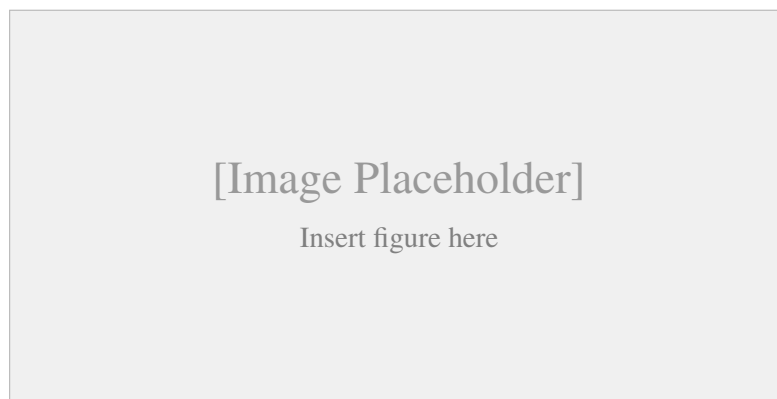


Figure 1.4: Electrolytic capacitor (radial type, 100 μF / 400 V)

Specifications. Capacitance values span an enormous range—from a few femtofarads (fF) for small ceramic chip capacitors to tens of thousands of microfarads (μF) for large electrolytic types. Key specifications include capacitance value, voltage rating (the maximum DC voltage that can be sustained without breakdown), temperature coefficient (how capacitance changes with temperature), equivalent series resistance (ESR), and leakage current. Common dielectric materials include ceramic (stable, low loss, suitable for high-frequency bypass), electrolytic oxide film (high capacitance density, polarized), tantalum oxide (compact, polarized, low ESR), and polyester film (non-polarized, general-purpose) [2].

Purpose of Use. Capacitors serve several distinct roles depending on the circuit context. They store energy and release it when needed (energy storage), block DC while passing AC signals (coupling and DC blocking), smooth out ripple in power supply outputs (filtering),

tune resonant circuits (in combination with inductors), and create time delays in RC timing circuits.

Applications. Capacitors are found in every electronic sub-system. Power supply filter capacitors reduce ripple on DC rails; bypass capacitors suppress high-frequency noise on power pins of ICs; coupling capacitors transfer audio or RF signals between stages without passing the DC bias; and timing capacitors set oscillation frequencies in clock circuits and astable multivibrators [9].

1.3.5 Diode

Definition. A diode is a two-terminal semiconductor device that allows electric current to flow predominantly in one direction, thereby functioning as a one-way valve for current [2]. It is formed by a junction of p-type and n-type semiconductor material (a p–n junction). When the anode is positive with respect to the cathode (forward bias), the diode conducts with a typical forward voltage drop of about 0.6–0.7 V for silicon; in the reverse direction (reverse bias), it blocks current until the reverse breakdown voltage is exceeded.

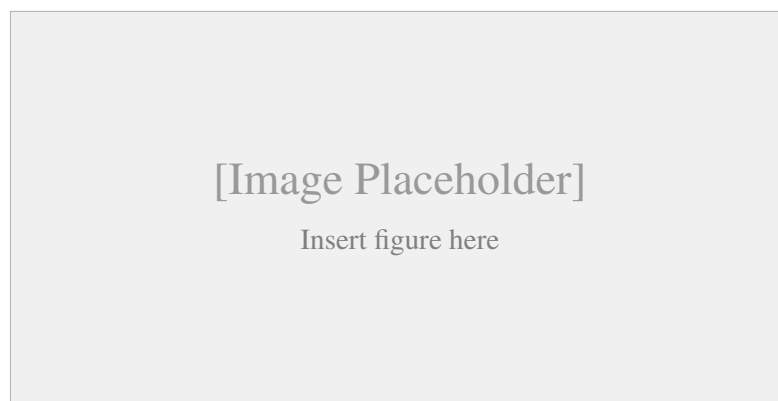


Figure 1.5: Rectifier diode (1N4003) showing physical package and schematic symbol

Specifications. Key diode specifications are the forward voltage drop (V_F), the maximum forward current (I_F), the peak inverse voltage (PIV) or reverse breakdown voltage, and the reverse saturation current (I_S). A standard rectifier diode such as the 1N4007 has $V_F \approx 1.1$ V at 1 A, PIV of 1000 V, and an average forward current rating of 1 A. Fast-recovery diodes and Schottky diodes offer lower V_F (0.2–0.4 V for Schottky) and faster switching times for high-frequency applications [1].

Purpose of Use. The primary use of a diode is rectification—converting AC to DC by allowing only one polarity of current to pass. Diodes are also used for voltage clamping (protecting circuits from voltage spikes), signal demodulation in radio receivers, logic level shifting, and, in the case of light-emitting diodes, the generation of visible or infrared light.

Applications. Diodes appear in AC–DC power supply rectifier bridges, battery chargers (to prevent reverse current), signal detectors in AM radio receivers, freewheeling or flyback circuits that protect switching transistors from inductive voltage spikes, and overvoltage protection circuits on the inputs of logic ICs [10].

1.3.6 Zener Diode

Definition. A Zener diode is a special type of silicon p–n junction diode designed to operate reliably in the reverse-bias breakdown region [2]. Unlike a conventional rectifier diode, which is destroyed by sustained reverse breakdown, the Zener diode is deliberately constructed so that once the applied reverse voltage reaches its characteristic *Zener voltage* (V_Z), it conducts in the reverse direction while maintaining that voltage nearly constant over a wide range of current. This phenomenon, first described by American physicist Dr. Clarence Melvin Zener, makes the device invaluable as a voltage reference.

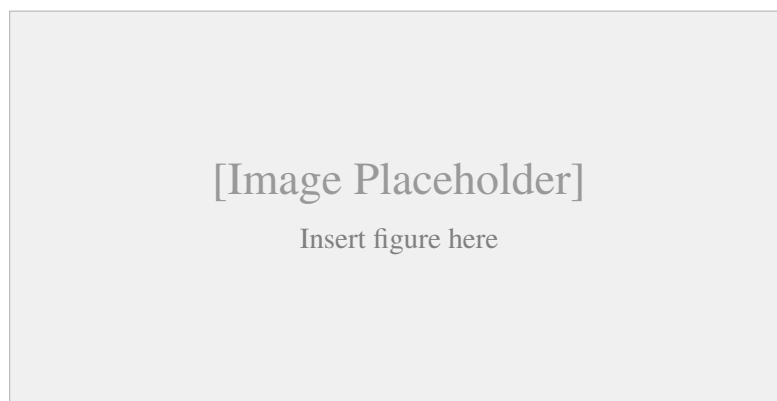


Figure 1.6: Zener diode in TO-92 package with schematic symbol

Specifications. The most important specification of a Zener diode is its Zener (breakdown) voltage V_Z , which is available in standard values ranging from about 2.4 V to 200 V. Additional parameters include the power dissipation rating (P_Z), the dynamic impedance (Z_Z , typically a few ohms, indicating how stable the voltage is under varying current), and the temperature coefficient of V_Z (which can be positive, negative, or near zero depending on V_Z) [1]. Common power ratings for small-signal Zener diodes are 400 mW and 1 W.

Purpose of Use. The Zener diode’s most fundamental use is voltage regulation—maintaining a stable output voltage from a supply that may vary in magnitude. When connected in reverse bias across a load with a series current-limiting resistor, the Zener clamps the output voltage to V_Z regardless of moderate variations in the input supply or load current.

Applications. Zener diodes provide voltage references in regulated power supplies, protect circuits from overvoltage transients (clamping), set threshold voltages in comparator circuits,

and stabilize bias voltages in analog amplifier stages [2]. They are found in smartphones, digital multimeters, surge-protected power strips, and battery management systems.

1.3.7 Light-Emitting Diode (LED)

Definition. A light-emitting diode (LED) is a semiconductor p–n junction device that emits light when forward-biased current flows through it [1]. As electrons from the n-type region recombine with holes in the p-type region, they release energy in the form of photons. The wavelength—and hence color—of the emitted light is determined by the bandgap energy of the semiconductor material used. LEDs are the most visible member of the diode family and are used both as indicators and as high-efficiency light sources.

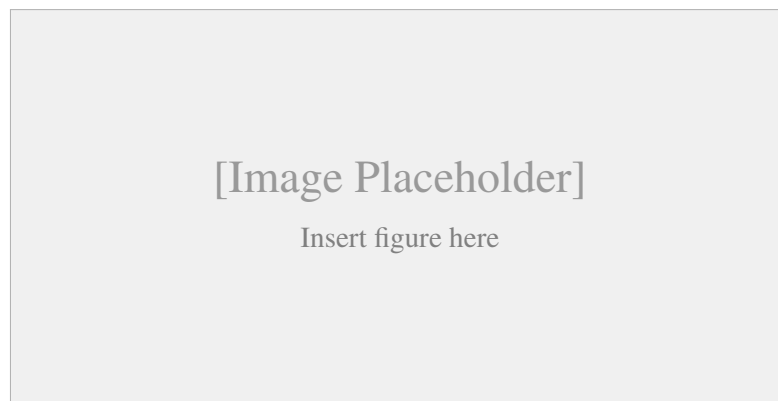


Figure 1.7: LED structure diagram showing anode (long lead) and cathode (short lead)

Specifications. Key LED specifications include the forward voltage V_F (typically 1.8–2.2 V for red or yellow, 3.0–3.5 V for blue and white), maximum continuous forward current I_F (commonly 20 mA for standard 5 mm indicator LEDs), luminous intensity (measured in millicandela, mcd), viewing angle (typically 15°–60°), and dominant wavelength. High-power LEDs for lighting applications carry current ratings of 350 mA to several amperes and are mounted on heatsinks [9].

Purpose of Use. In the laboratory context, LEDs are most commonly used as circuit-state indicators—visual confirmation that a circuit or a particular branch is energized. They must always be paired with a current-limiting resistor ($R = (V_{supply} - V_F)/I_F$) to prevent destructive over-current.

Applications. LEDs are ubiquitous in electronic displays, indicator lights, backlighting for LCD screens, automotive lighting (headlamps and tail lights), traffic signals, optical fiber communications (infrared LEDs), remote controls, and solid-state general-purpose lighting. Their energy efficiency—far superior to incandescent bulbs—has made them the dominant light source in modern electronics [11].

1.3.8 Veroboard

Definition. A veroboard (also called a stripboard) is a type of electronics prototyping board constructed from a single-sided, copper-clad, fiberglass-reinforced epoxy laminate (FR4) with a regular matrix of holes [7]. Unlike a breadboard, a veroboard is intended for semi-permanent circuits: components are soldered in place, and the continuous copper tracks (strips) running between rows of holes provide electrical connections. Where connections need to be broken, a track-cutter tool is used to isolate individual sections.

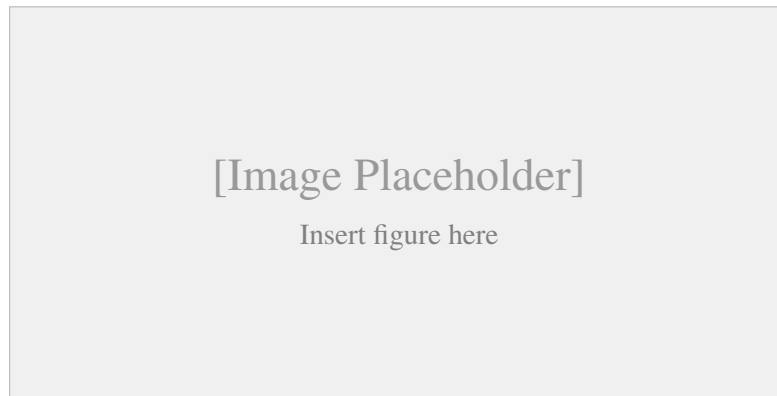


Figure 1.8: Veroboard (stripboard) showing copper tracks on the underside

Specifications. Standard veroboard has a hole pitch of 2.54 mm (0.1 inch), compatible with DIP integrated circuit packages. The copper strip width is typically 1.5 mm with 0.5 mm gaps between adjacent strips. Board thickness is usually 1.6 mm (FR4) and the copper foil weight is 35 μm (1 oz). Unlike breadboards, veroboards can safely handle continuous currents of several amperes through the copper strips and are suitable for audio-frequency and low RF circuits where the soldered connections provide reliable electrical contact [8].

Purpose of Use. Veroboard bridges the gap between breadboard prototyping and custom PCB manufacture. Once a circuit has been verified on a breadboard, it can be transferred to a veroboard for a more robust, semi-permanent realization. Component placement is planned on paper or with CAD tools, taking care to plan the track cuts needed to isolate nodes.

Applications. Veroboards are used in electronics hobbyist projects, small-batch production of simple circuits, repair workshops (where custom PCBs are not economical), audio amplifier construction, and educational laboratories where students need a permanent record of their circuit work [2].

1.3.9 Oscilloscope

Definition. An oscilloscope is an electronic test instrument that graphically displays the variation of a voltage signal over time [12]. It plots voltage on the vertical (Y) axis against

time on the horizontal (X) axis, revealing the shape—and therefore the key parameters—of an electrical signal. Modern digital storage oscilloscopes (DSOs) sample the analog input, digitize it with an analog-to-digital converter, and store the waveform in memory for display and analysis.

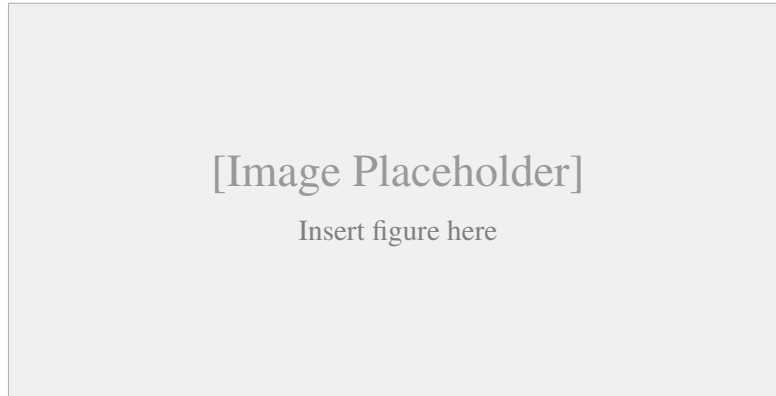


Figure 1.9: Digital storage oscilloscope displaying a sinusoidal waveform

Specifications. The most critical oscilloscope specification is its *bandwidth*—the frequency at which a sinusoidal input signal is attenuated by 3 dB (reduced to 70.7% of its true amplitude) [13]. Laboratory teaching oscilloscopes commonly have bandwidths of 50–200 MHz; professional engineering instruments may span 1 GHz to over 100 GHz. The *sample rate* (in samples per second, S/s) must be at least twice the highest signal frequency (Nyquist criterion); modern DSOs offer 1 GS/s to over 100 GS/s. Additional specifications include memory depth (number of samples stored per acquisition), vertical sensitivity (typically 1 mV/div to 10 V/div), number of input channels, and trigger options [14].

Purpose of Use. The oscilloscope allows the engineer to observe signal waveforms that a multimeter cannot capture, such as rapid transients, oscillation, noise, and timing relationships between two or more signals. It is essential for verifying the shape of filter outputs, the rise time of digital pulses, the duty cycle of PWM signals, and the correct timing of serial communication protocols.

Applications. Oscilloscopes are used in electronics design and debug labs, manufacturing test, radio and telecommunications engineering, power electronics (measuring switching transients in inverters and converters), automotive electronics (decoding CAN bus messages), and biomedical instrumentation (displaying ECG, EEG, and other biopotential signals) [12].

1.3.10 Operational Amplifier (Op-Amp)

Definition. An operational amplifier (op-amp) is a high-gain, DC-coupled differential voltage amplifier packaged as an integrated circuit [15]. It has two inputs—an inverting input (−) and a non-inverting input (+)—and one output. Its *open-loop* gain (the gain without

any external feedback network) is very large: ideally infinite, and practically in the range of 20,000 to 200,000. In virtually all practical applications, negative feedback is applied via external resistors or other components to set a precise, stable closed-loop gain.

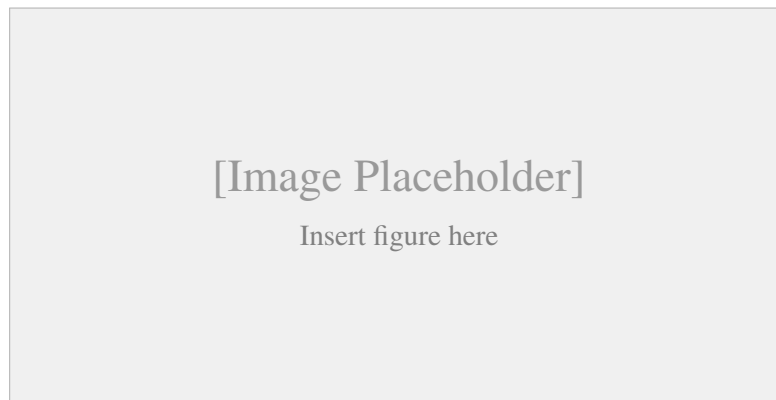


Figure 1.10: LM741 op-amp IC (DIP-8 package) with pin diagram and inverting amplifier circuit

Specifications. For an ideal op-amp: open-loop gain $A_{OL} = \infty$, input impedance $Z_{in} = \infty$, output impedance $Z_{out} = 0$, bandwidth $= \infty$, and input offset voltage $= 0$ [16]. A practical general-purpose op-amp such as the LM741 exhibits $A_{OL} \approx 200,000$, $Z_{in} \approx 2\text{ M}\Omega$, $Z_{out} \approx 75\ \Omega$, a unity-gain bandwidth of about 1 MHz, and an input offset voltage of 1–6 mV. Other key parameters include the slew rate (maximum rate of output voltage change, e.g. $0.5\text{ V}/\mu\text{s}$ for the LM741), common-mode rejection ratio (CMRR, typically 70–90 dB), and supply voltage range [9].

Purpose of Use. Op-amps are used to amplify weak signals to useful levels, to implement mathematical operations (summation, integration, differentiation—whence the name *operational* amplifier), to compare two voltages, to convert current to voltage, and to buffer a high-impedance source to drive a low-impedance load without loading the source.

Applications. Op-amps are found in audio preamplifiers and equalizers, active filter circuits (low-pass, high-pass, bandpass), instrumentation amplifiers for sensor interfaces (thermocouples, load cells, strain gauges), analog-to-digital converter front-ends, signal generators (Wien bridge oscillators, function generators), and feedback control systems in industrial automation and robotics [1].

1.3.11 Bipolar Junction Transistor (BJT)

Definition. A bipolar junction transistor (BJT) is a three-terminal semiconductor device that uses both electrons and holes as charge carriers—hence the term *bipolar* [2]. It consists of two back-to-back p–n junctions sharing a thin common region called the *base*, flanked by the *emitter* and *collector* regions. BJTs are manufactured in two complementary configurations: NPN (n-type emitter and collector surrounding a p-type base) and PNP (p-type emitter and

collector surrounding an n-type base). A small current injected into the base controls a much larger collector–emitter current, making the BJT a current-controlled device.

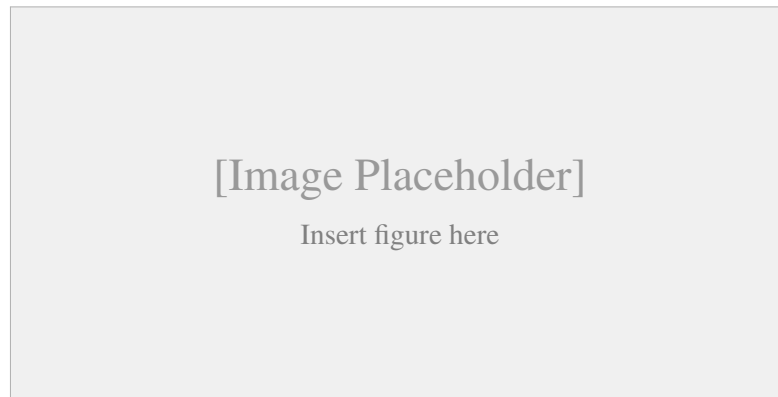


Figure 1.11: NPN BJT schematic symbol, terminal labels, and TO-92 package

Specifications. The primary BJT specification is the DC current gain h_{FE} (or β), the ratio of collector current to base current, typically ranging from 20 to several hundred depending on device type and operating conditions. Other key parameters include the collector–emitter breakdown voltage (V_{CEO}), the collector–emitter saturation voltage ($V_{CE(sat)}$ for switching applications), the maximum collector current I_C , the maximum power dissipation P_C , and the transition frequency f_T (the frequency at which h_{FE} drops to unity) [1].

Purpose of Use. BJTs are used in two fundamental operating modes. In the *active region* (base–emitter junction forward biased, base–collector junction reverse biased), the BJT acts as a linear current amplifier. In the *saturation/cutoff region* (both junctions forward or reverse biased, respectively), it acts as an electronic switch. The choice between NPN and PNP depends on the polarity of the supply and the circuit topology required.

Applications. BJTs are used in audio amplifiers (common-emitter and push-pull output stages), radio frequency amplifiers, oscillators, logic circuits (TTL gates), motor drivers, relay drivers, temperature sensors (the V_{BE} of a BJT is predictably temperature-dependent), and power regulators [11]. In modern mixed-signal integrated circuits, BJTs coexist with CMOS transistors in BiCMOS processes to exploit the high-speed characteristics of bipolar devices alongside the low power of CMOS.

1.3.12 Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

Definition. A MOSFET (metal-oxide-semiconductor field-effect transistor) is a three-terminal semiconductor device in which the current between the *drain* and *source* terminals is controlled by the voltage applied to the insulated *gate* terminal [17]. The gate is separated from the semiconductor channel by a thin layer of silicon dioxide (SiO_2), making it electrically insulated; consequently, virtually no gate current flows, giving the MOSFET a very high input

impedance. MOSFETs are fabricated as n-channel (NMOS) or p-channel (PMOS) types; complementary pairs (CMOS) are the basis of modern digital logic.

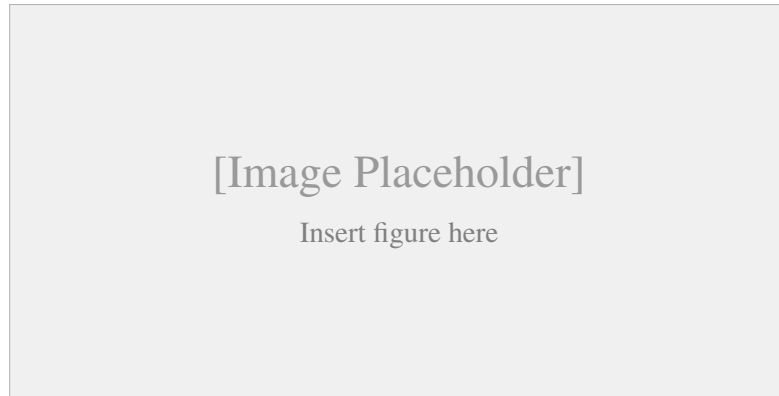


Figure 1.12: N-channel enhancement MOSFET in TO-220 package with schematic symbol and pin layout

Specifications. Key MOSFET parameters include the gate threshold voltage $V_{GS(th)}$ (the minimum gate-to-source voltage required to form a conductive channel), the drain-to-source on-state resistance $R_{DS(on)}$ (a few milliohms for power MOSFETs, directly affecting conduction losses), maximum drain current I_D , drain-to-source breakdown voltage V_{DSS} , and transconductance g_m [11]. For power switching applications, switching speed (rise/fall times of a few nanoseconds to tens of nanoseconds) and the gate charge Q_g (determining how quickly the gate can be driven) are also critical.

Purpose of Use. In the saturation region, the MOSFET acts as a voltage-controlled current source suitable for linear amplification. In the cut-off and triode (ohmic) regions, it acts as an electronic switch—the dominant use in digital and power electronics. Because MOSFETs switch 50 to 100 times faster than comparable BJTs, they are the preferred choice in high-frequency power converters [17].

Applications. MOSFETs are the most widely manufactured semiconductor devices in the world. In digital integrated circuits, billions of CMOS MOSFETs constitute the logic gates, memory cells, and processors in every computer and smartphone. In power electronics, discrete power MOSFETs switch thousands of watts in switch-mode power supplies, DC motor drives, solar inverters, and battery management systems. RF power MOSFETs amplify signals in wireless base stations, and audio MOSFETs drive high-fidelity loudspeakers [17].

1.3.13 Soldering Iron

Definition. A soldering iron is a hand-held tool with an electrically heated metal tip, called the *bit*, used to melt solder and join metal parts, wires, or electronic component leads to circuit boards [18]. Solder is a low-melting-point metal alloy (traditionally 60/40 tin–lead,

melting at approximately 183 °C; increasingly lead-free alloys such as Sn–Ag–Cu melting at 217–227 °C) that, when melted and allowed to solidify, forms a mechanically strong and electrically conductive joint.

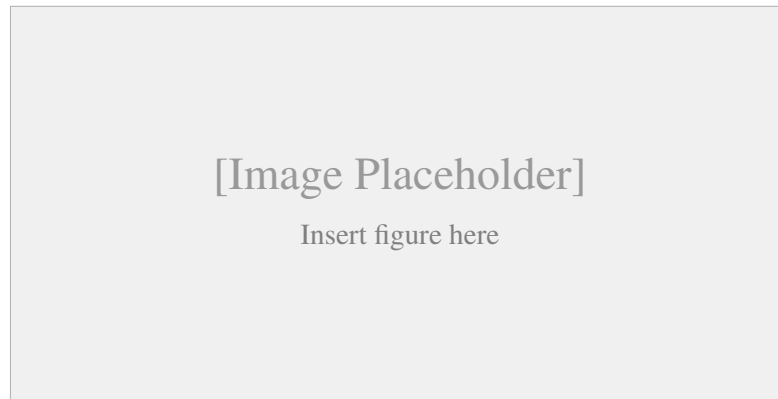


Figure 1.13: Electrically heated soldering iron with replaceable bit and insulated handle

Specifications. Soldering irons are designed to operate within a temperature range of approximately 200 °C to 480 °C (392–896 °F) [18]. For general electronics work with lead-free solder, a tip temperature of 315–370 °C is typical; lower temperatures (300 °C or below) are used for heat-sensitive SMD components. Power ratings range from about 15 W for lightweight precision irons to 60–100 W for heavy-duty or soldering station units. Temperature-controlled soldering stations include a thermocouple or Curie-point sensor at the tip and closed-loop control circuitry to maintain the set temperature within a few degrees [18].

Purpose of Use. The soldering iron’s purpose is to create permanent, reliable electrical connections by reflowing solder onto component leads and PCB pads. Good solder joints exhibit a smooth, shiny appearance and low contact resistance; a poorly made *cold joint* is dull and grainy, with unreliable conductivity. The soldering iron is also used to desolder components for rework, using either a desoldering pump or wick.

Applications. Soldering irons are used in electronics assembly and repair workshops, PCB prototyping on veroboard, rework and modification of existing PCBs, installation of connectors and terminal blocks, hobbyist projects, and educational laboratories. High-volume production lines replace manual soldering with automated wave or reflow soldering ovens, but a soldering iron remains essential for rework and repair tasks that cannot be automated [18].

1.4 Procedure

1. Observe each electronic component and instrument demonstrated in the laboratory.
2. Carefully examine each component by hand, noting its physical characteristics (size, number of terminals, marking, color bands, package type).

3. Note down the IC or device part number where applicable.

4. Understand the working principle and important features of each component as described in the preceding theory sections.

1.5 Discussion

1.6 Conclusion

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